Customer Behavior Analysis Report

1. Introduction

This project focuses on analyzing customer transaction data to understand customer behavior, identify different customer segments, analyze purchase patterns, and detect churn risks. The goal is to provide actionable insights that can help improve customer engagement and business performance.

2. Data Cleaning

The dataset was cleaned to ensure accuracy and consistency. Missing values were removed, and duplicate records were eliminated. The purchase date column was converted into a proper datetime format to enable time-based analysis.

3. Feature Engineering

New features were created to enhance analysis:

- Extracted **Year** and **Month** from the purchase date.
- Used **Total Purchase Amount** to analyze customer spending behavior.
- Removed redundant columns to improve data quality.

4. Customer Segmentation (RFM Analysis)

Customers were segmented using the RFM model:

- **Recency (R):** Days since last purchase

- **Frequency (F):** Number of purchases
- **Monetary (M):** Total amount spent

Based on this, customers were classified into:

- **Low Value Customers:** Low frequency and spending
- **Regular Customers:** Moderate purchase behavior
- **High Value Customers:** Frequent buyers with high spending

This segmentation helps in understanding customer importance and targeting them effectively.

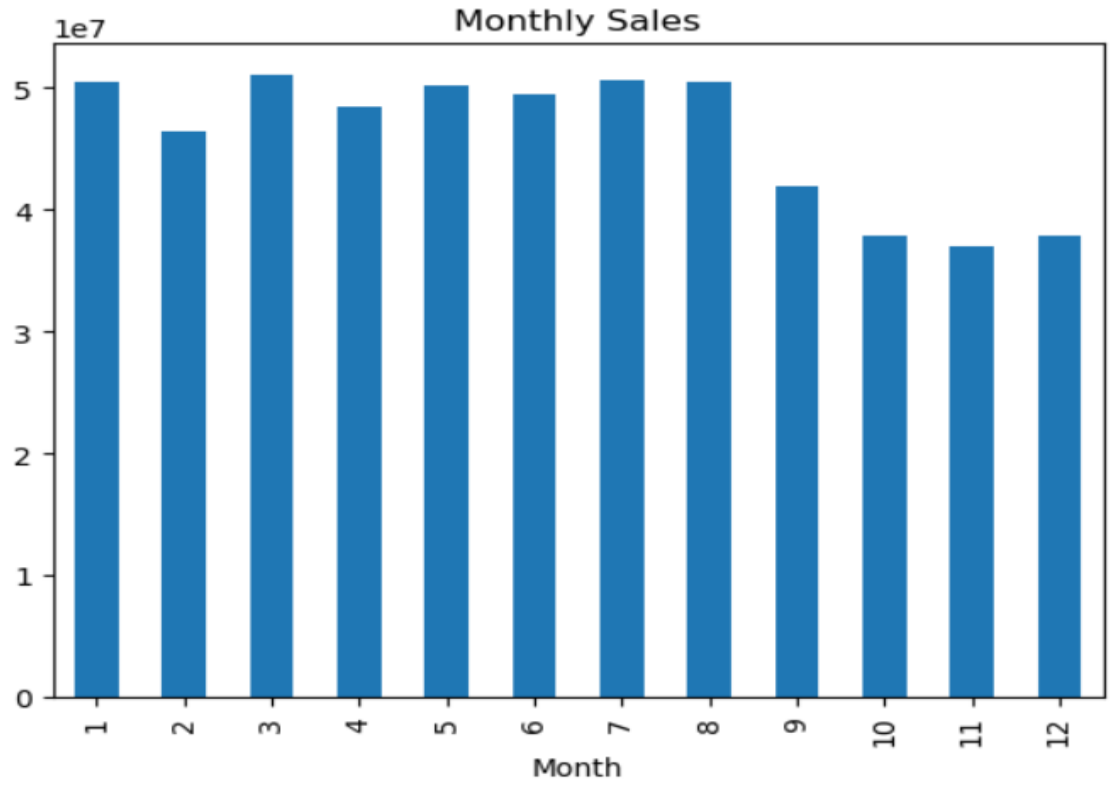
5. Data Visualization

Several visualizations were created to identify trends:

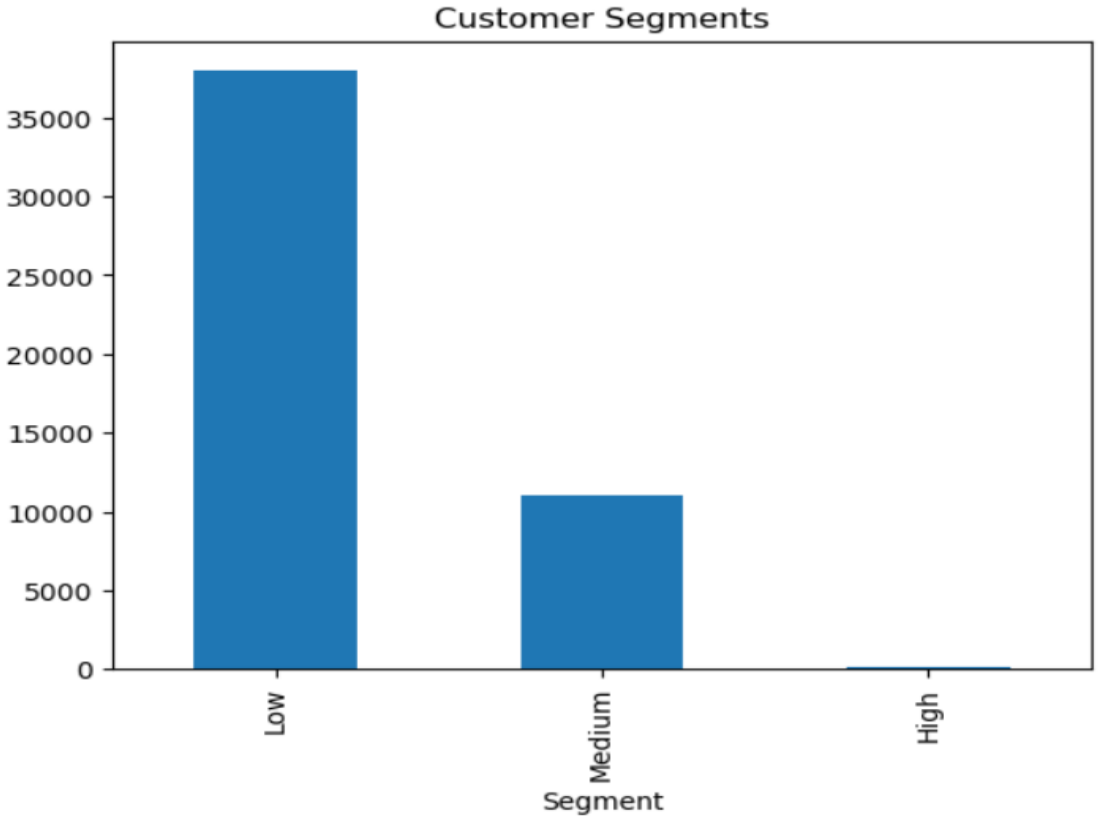
- **Monthly Sales Trend:** Showed variation in sales across months.
- **Customer Segments Distribution:** Displayed the number of customers in each segment.
- **Churn Distribution:** Highlighted the number of customers who stopped purchasing.
- **Product Category Distribution:** Showed which product categories are most popular.

These visualizations helped in identifying purchase patterns and customer behavior.

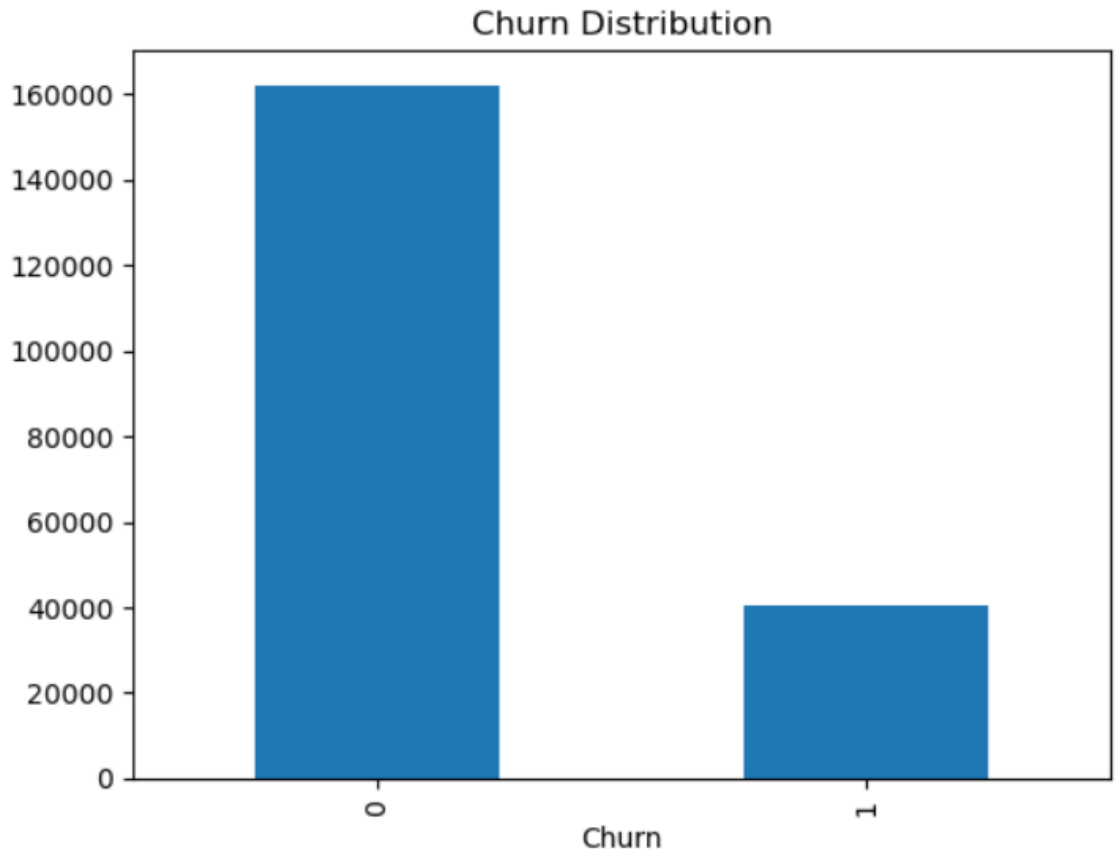
5.1 Monthly Sales Trend



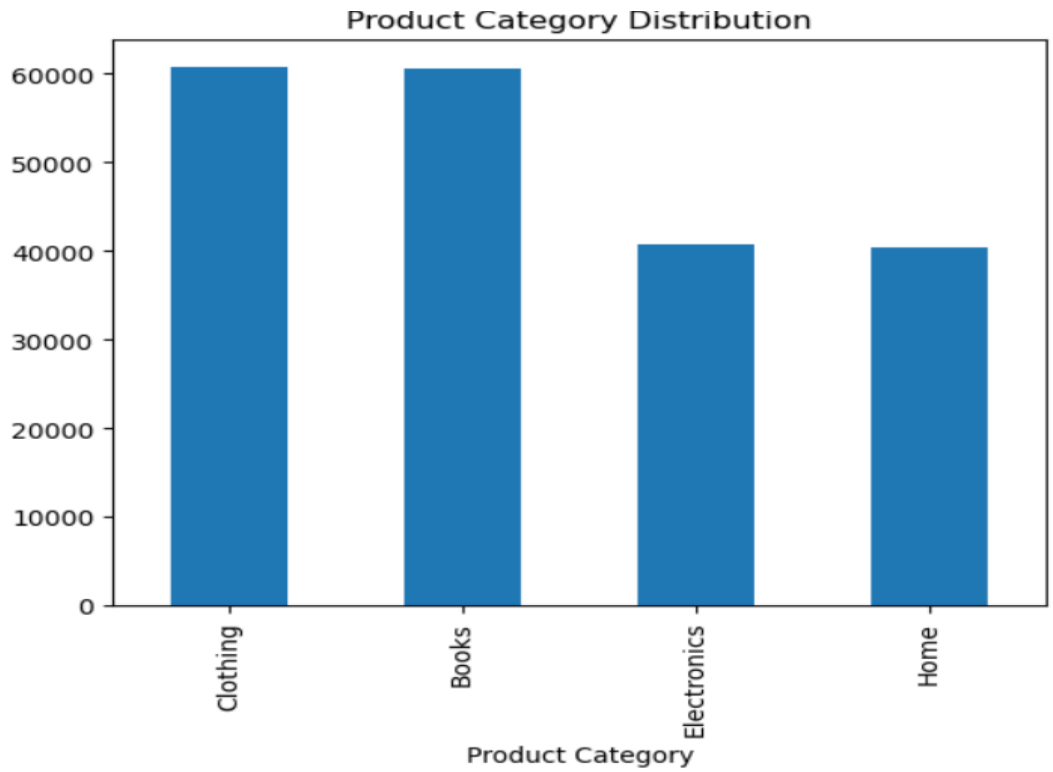
5.2 Customer Segments Distribution



5.3 Churn Distribution



5.4 Product Category Distribution



6. Key Insights

- A large number of customers fall into the low-value category.
- A small group of customers contributes significantly to total revenue.
- Sales vary across different months, indicating seasonal trends.
- Some customers are at risk of churn, which can affect business growth.
- Certain product categories perform better than others.

7. Recommendations

Based on the analysis, the following actions are recommended:

1. Offer discounts and promotions to low-value customers to increase engagement.
2. Introduce loyalty programs to retain high-value customers.
3. Target churn-risk customers with personalized offers and reminders.
4. Focus marketing efforts on high-performing months to maximize sales.
5. Personalize product recommendations based on customer preferences and behavior.

8. Conclusion

This analysis provided valuable insights into customer behavior, purchase patterns, and churn risks. By implementing the recommended strategies, Alfido Tech can improve customer engagement, increase retention, and enhance overall business performance.